



Steer-by-Wire: Modification of Vehicle Handling Characteristics

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Introduction

- A control strategy of the active front wheel steering system to protect from spin and to realize the improved cornering performance
“Bifurcation in Vehicle Dynamics and Robust Front Wheel Steering Control” by Eiichi Ono, Shigeyuki Hosoe, Hoang D. Tuan, and Shun’ichi Doi.
- Acceptable steering feel after the elimination of the link between steering wheel and road wheels.
“A Control System Methodology for Steer-by-Wire Systems” by Sanket Amberkar, Farhad Bolourchi, Jon Demerly and Scott Millsap.
- *Experimental Determination of Transfer Functions*
“Modern Control Engineering” by Katsuhiko Ogata.



Introduction (Cont.)

- Develop a procedure for the parameter identification of a steering system, processing experimental measurements obtained on a test bench.

“Identification of steering system parameters by experimental measurements processing” by S Data, M Pesce and L Reccia

- Design and implementation of a steer-by-wire system that provides active steering.

“Modification of Vehicle Handling Characteristics via Steer-by-Wire” by Paul Yih and J. Christian Gerdes

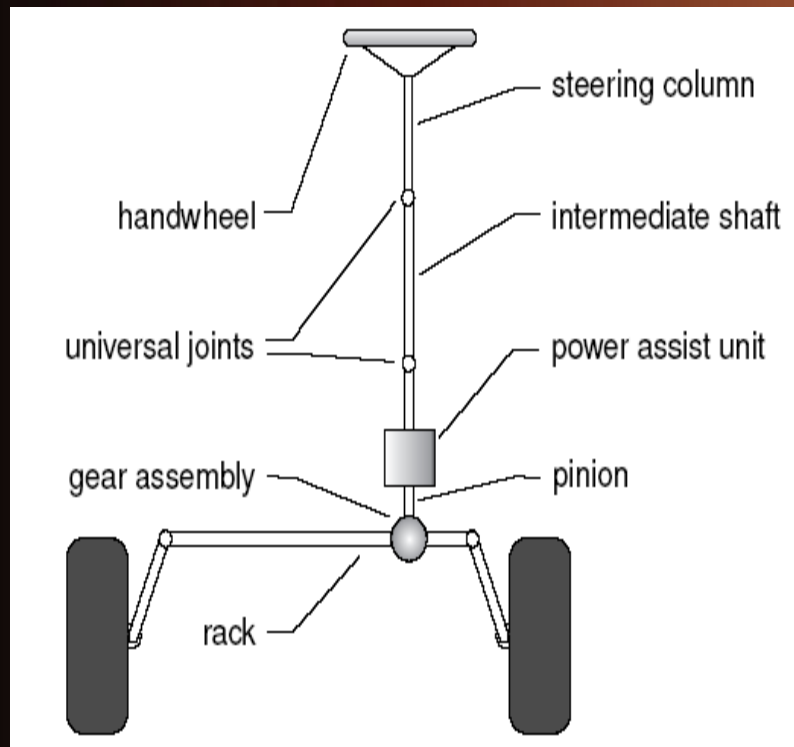


What is Steer-by-Wire?

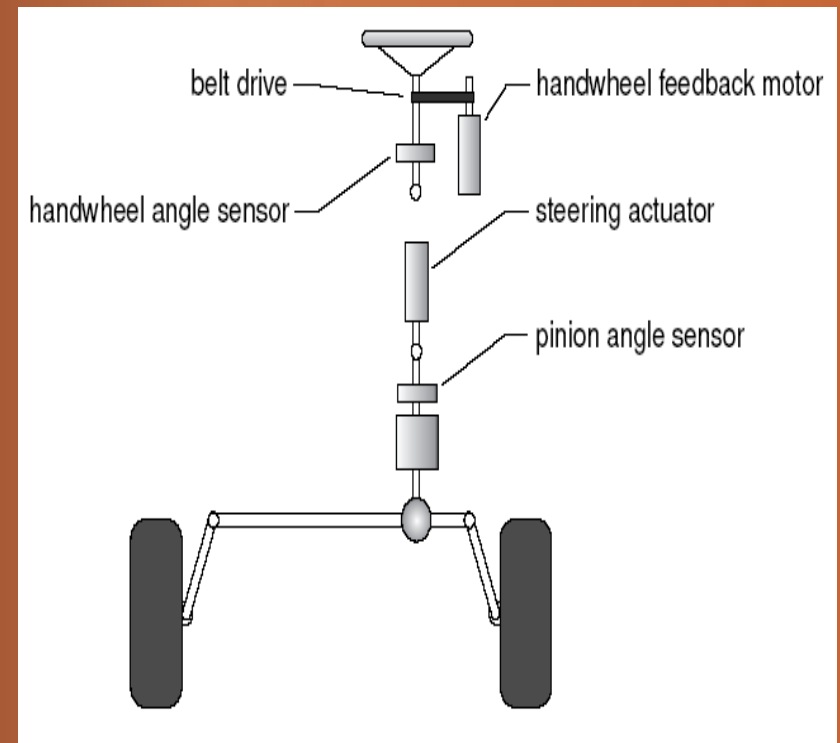
Unlike the conventional steering system where a hand-operated steering wheel is used to turn the front wheels through the steering column, steer-by-wire technology removes the mechanical and physical links between the driver (steering wheel) and the front wheels, and replace them with electronic actuators and other components.



Conventional Steering System



Steer-by-Wire System





Many Advantages

- No steering column – Simplify the design of a car's interior, giving the driver more space as well as better safety in case of a crash (no intrusion of the steering column).
- The absence of steering shaft and gear reduction mechanism allows much better utilization of the engine's compartment.
- Decreases the total weight of the car issuing better energy reduction effectiveness.
- Easier implementation of left or right-hand driving.
- No noise or vibration can reach the driver's hands.
- The most significant benefit is the ability to electronically augment the driver's steering input depending of drive's conditions, also called active steering.



The Objectives

- Simulate a conventional steering system
- Design and replace with active steering by-wire system
- New system must be comparable in response from a driver's perspective



The Experimental Transfer Function

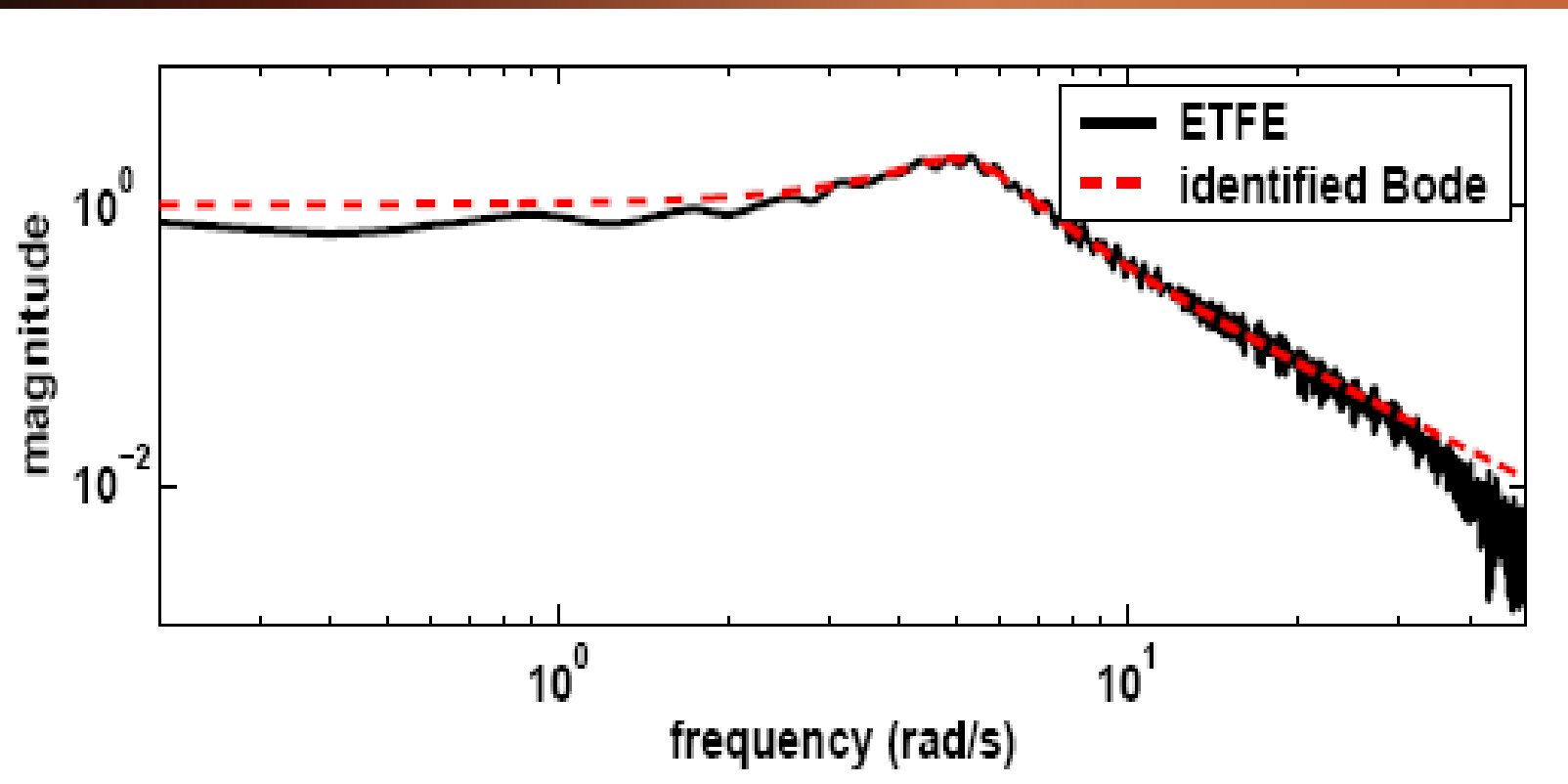
We need to find the transfer function without any tire force first, hence the front tires are off the ground. We accomplished that by looking at the magnitude and phase bode plot of the design article. Using asymptotic analysis of those two plots, we were able to determine the natural frequency, the damping ratio and the settling time of our system. With those values in hand, it was easy enough to determine the principal characteristics of the system such as the effective damping coefficient, the total moment of inertia and the gain of the steering system using the equivalence formula below:

$$\frac{\Theta(s)}{\Theta_d(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} = \frac{K}{J_s s^2 + b_s s + K}$$

where K is the gain, J is the moment of inertia and b is the damping coefficient.

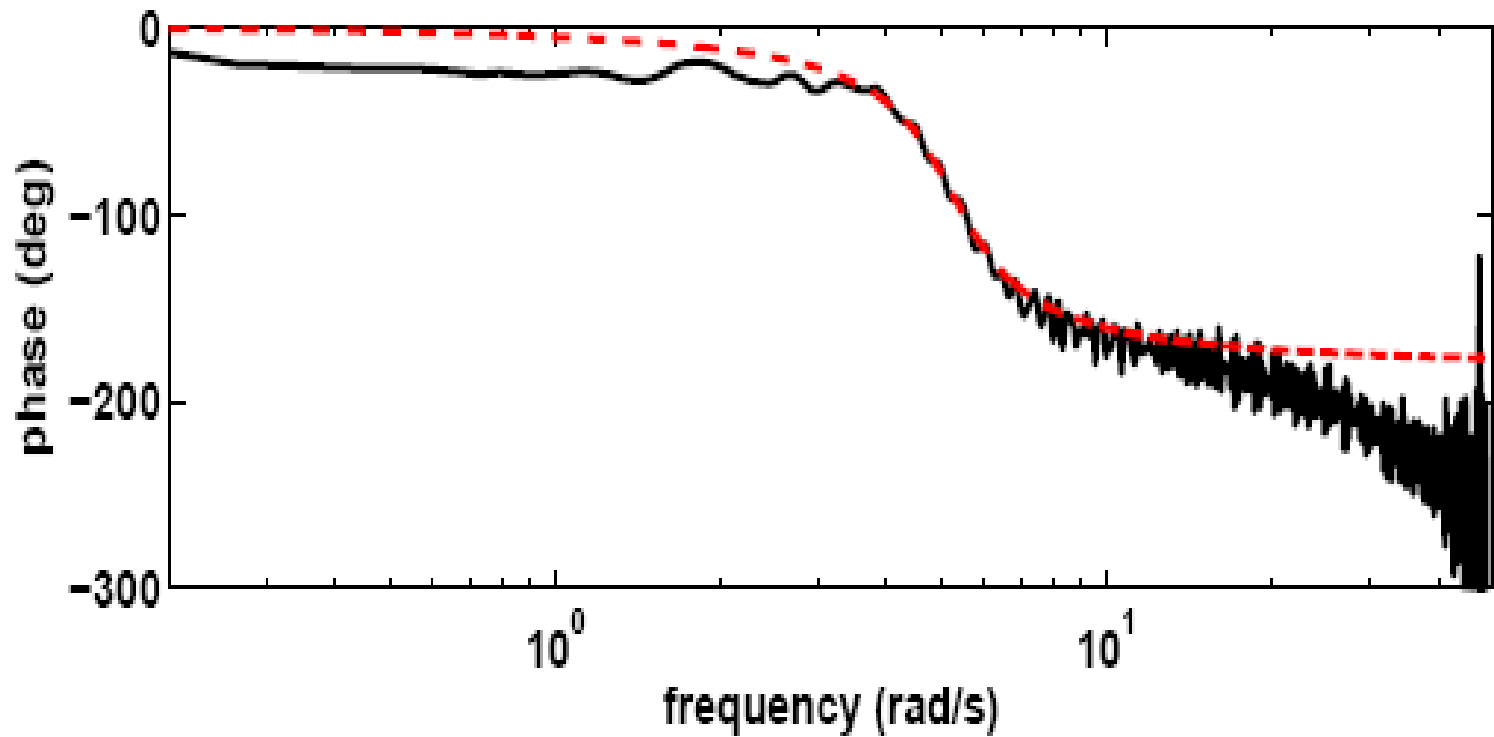


Magnitude Bode Plot



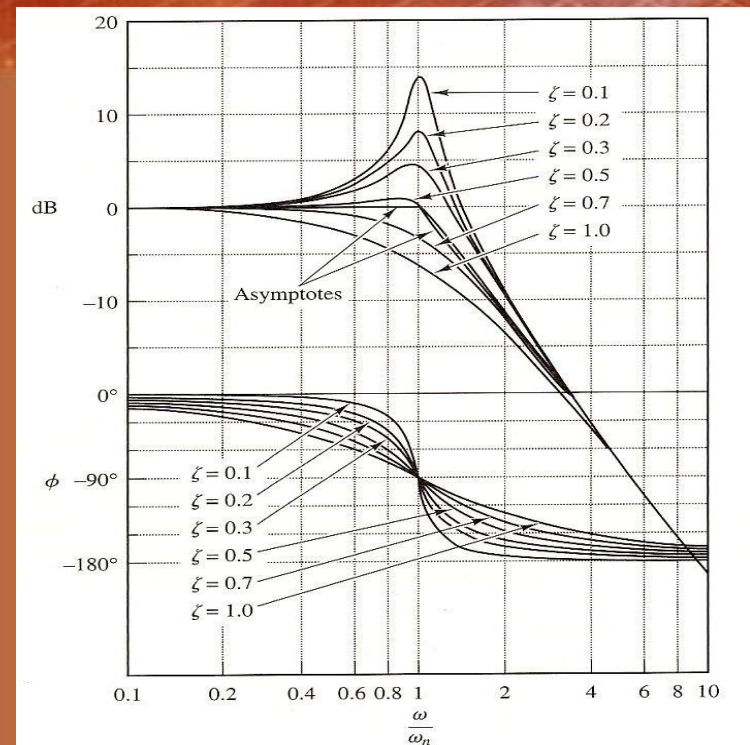


Phase Bode Plot





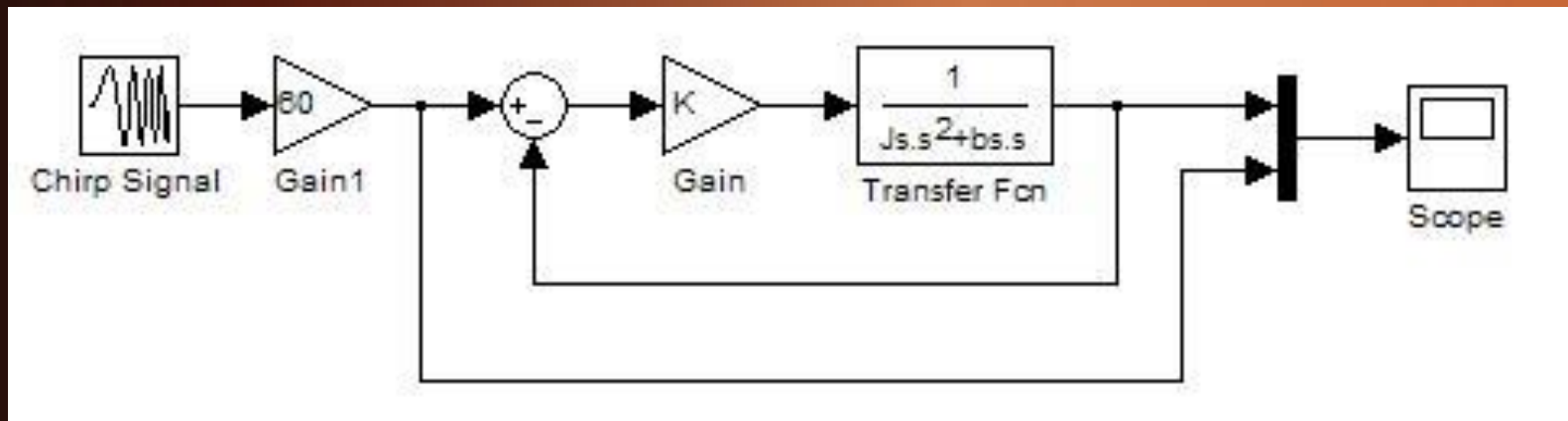
We notice on the magnitude bode plot that we have complex pole at $\omega_n = 4$ rad/sec. With only this information in hand, we already know that we have a system of 2nd order or more. Looking at the overshoot, we were also able to determine the damping ratio with the graph to the right. Finally, we determined the transfer function of the steering system:



$$\frac{\Theta(s)}{\Theta_d(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} = \frac{K}{J_s s^2 + b_s s + K}$$

$$= \frac{16}{s^2 + 2.4s + 16}$$

Simplified Design



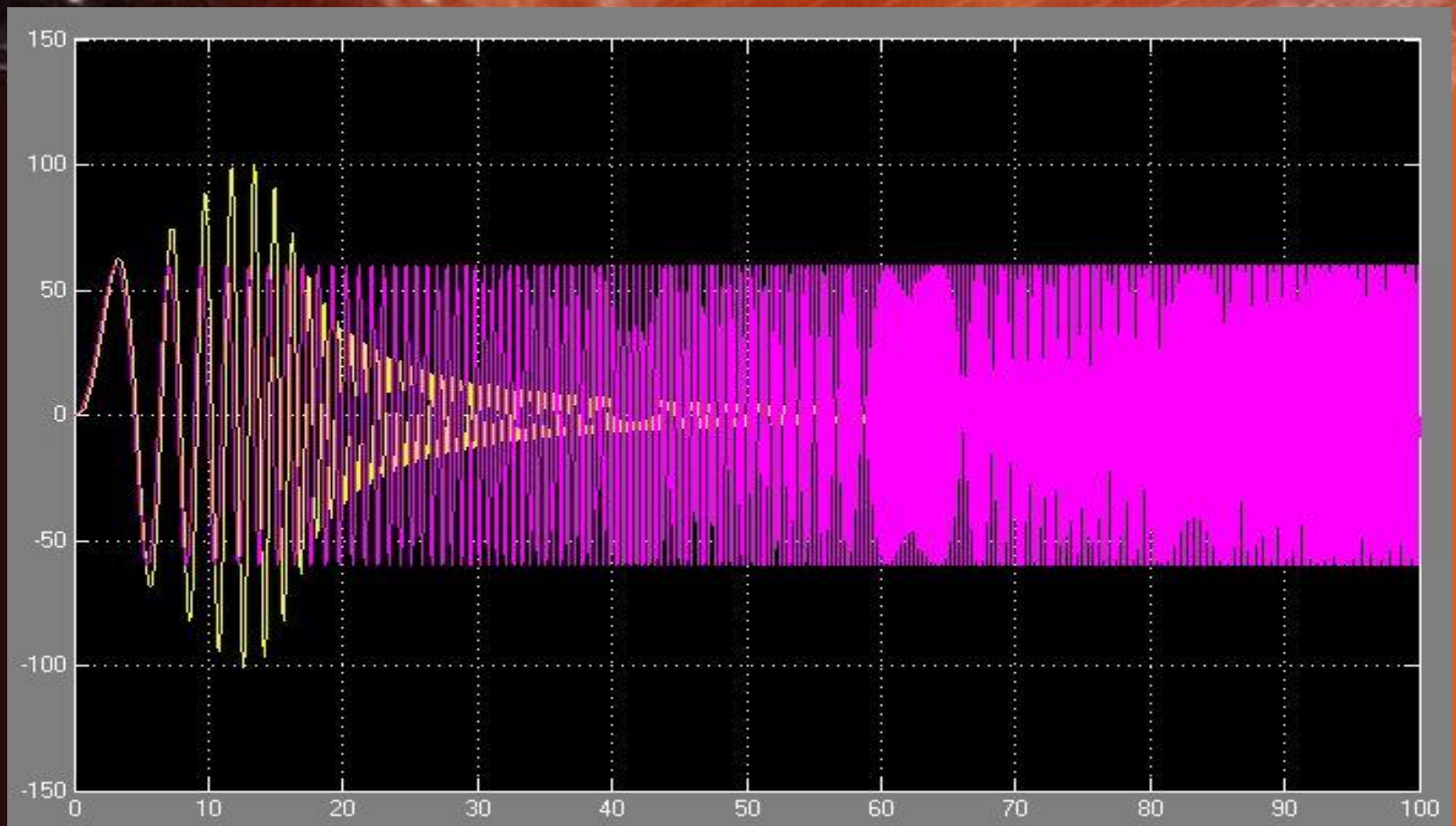
θ : pinion angle

τ : actuator torque

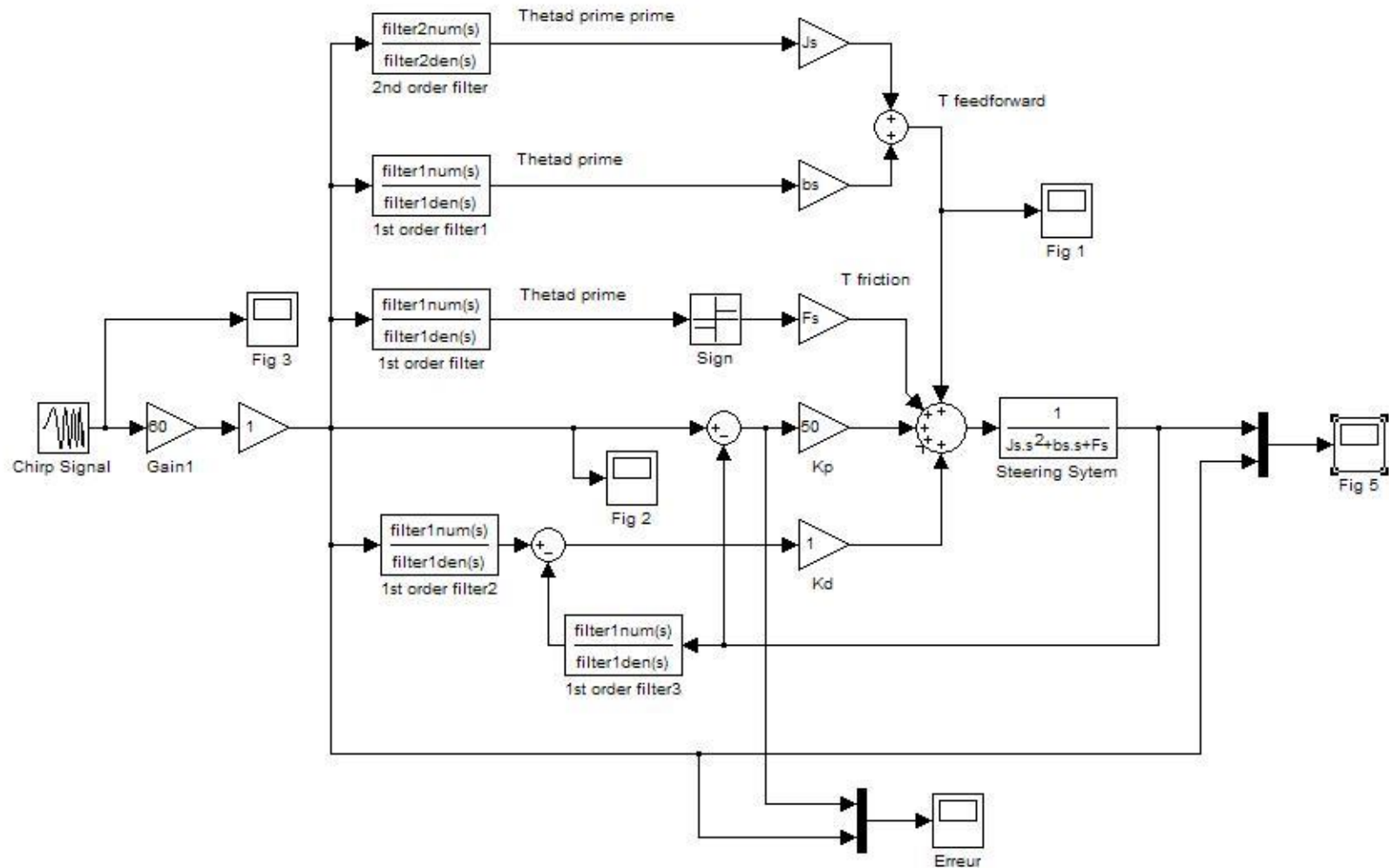
$$G(s) = \frac{\Theta(s)}{T(s)} = \frac{1}{J_s s^2 + b_s s}$$



Simplified Result



Steering System with Friction and Compensation





Design Components

- First order filter
- Second order filter
- Steering System

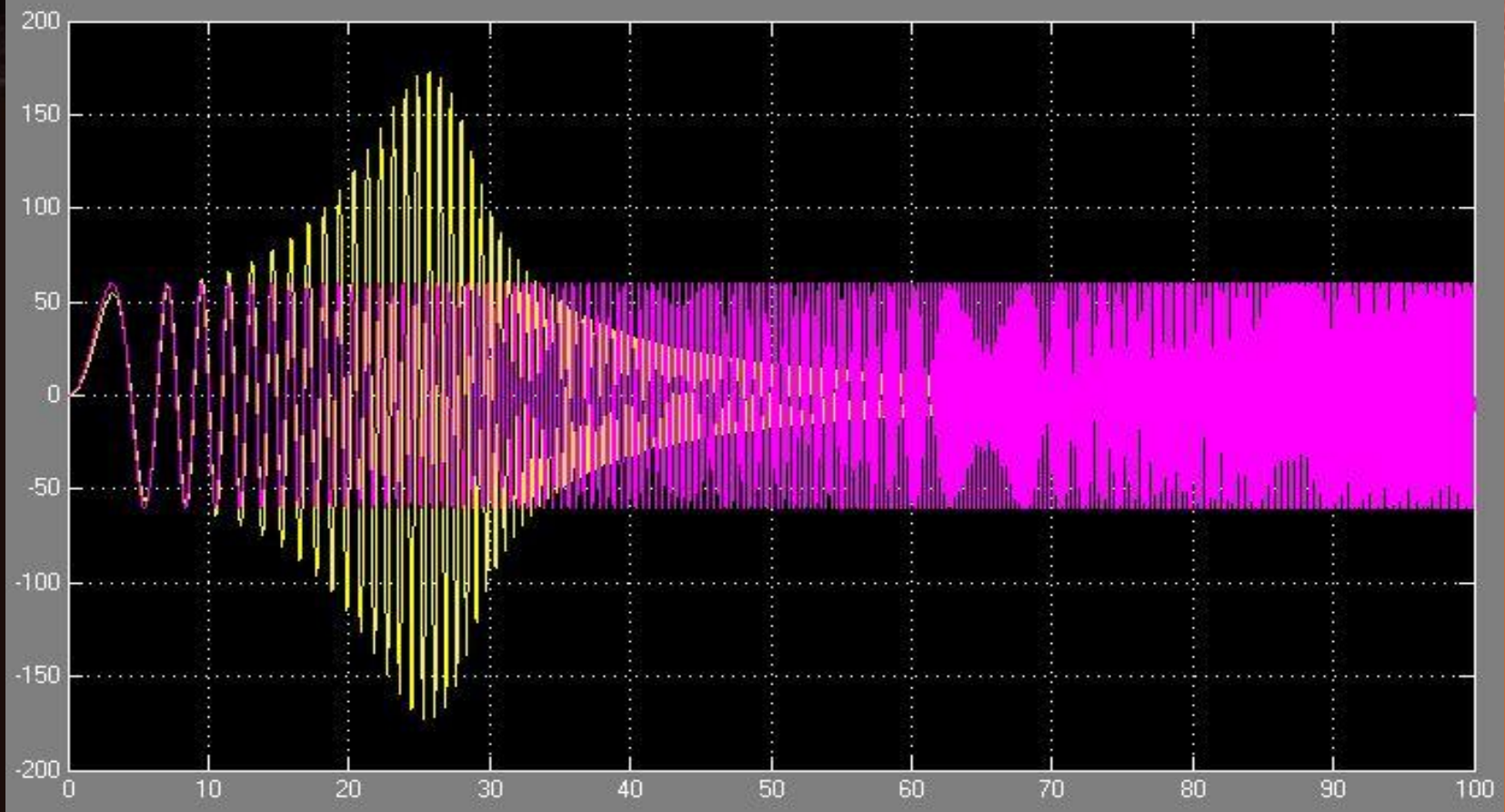
$$\frac{\Theta_d(s)}{\Theta_{d,sensor}(s)} = \frac{\omega_c}{s + \omega_c}$$

$$\frac{\Theta_d(s)}{\Theta_{d,sensor}(s)} = \frac{\omega_c^2}{s^2 + 2\omega_c s + \omega_c^2}$$

$$J_s \ddot{\theta} + b_s \dot{\theta} + F_s \operatorname{sgn}(\dot{\theta}) = \tau$$



Steering System Result





PD Controller

- We set the Proportional Gain **K_p to 50** and the Derivative Gain **K_d to 1** in order to achieve the same system response as the article paper. However, there is a reasoning behind those values.
- The high K_p will in fact increase the rise time of our system, which is the most important parameter of the system since we want the steering to act very quickly. The Derivative Gain, although very small, will help reducing the overshoot and the settling time of our system, without influencing the rise time by a lot.



Effect of tire self-aligning moment

- Consider tire-to-ground contact

- Total aligning moment: $\tau_a = (t_p + t_m)F_{y,f}(\alpha_f)$.

$F_{y,f}$: lateral force acting on the tire

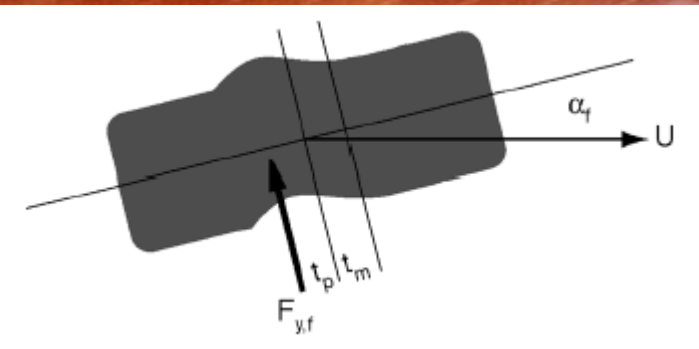
α_f : tire slip angle

t_p : pneumatic trail, the distance between the application of lateral force and the center of the tire

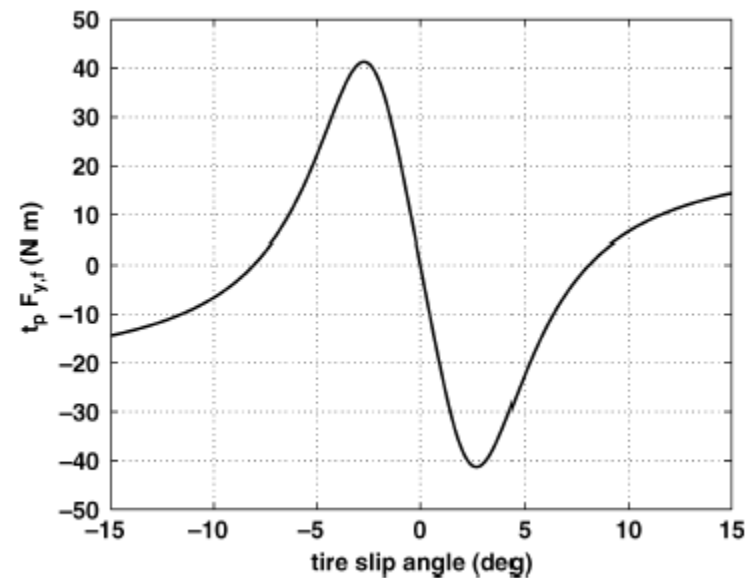
t_m : mechanical trail, the distance between the tire center and the ground



Tire operating at a slip angle:



Slip angle vs
component of aligning
moment due to
pneumatic trail:

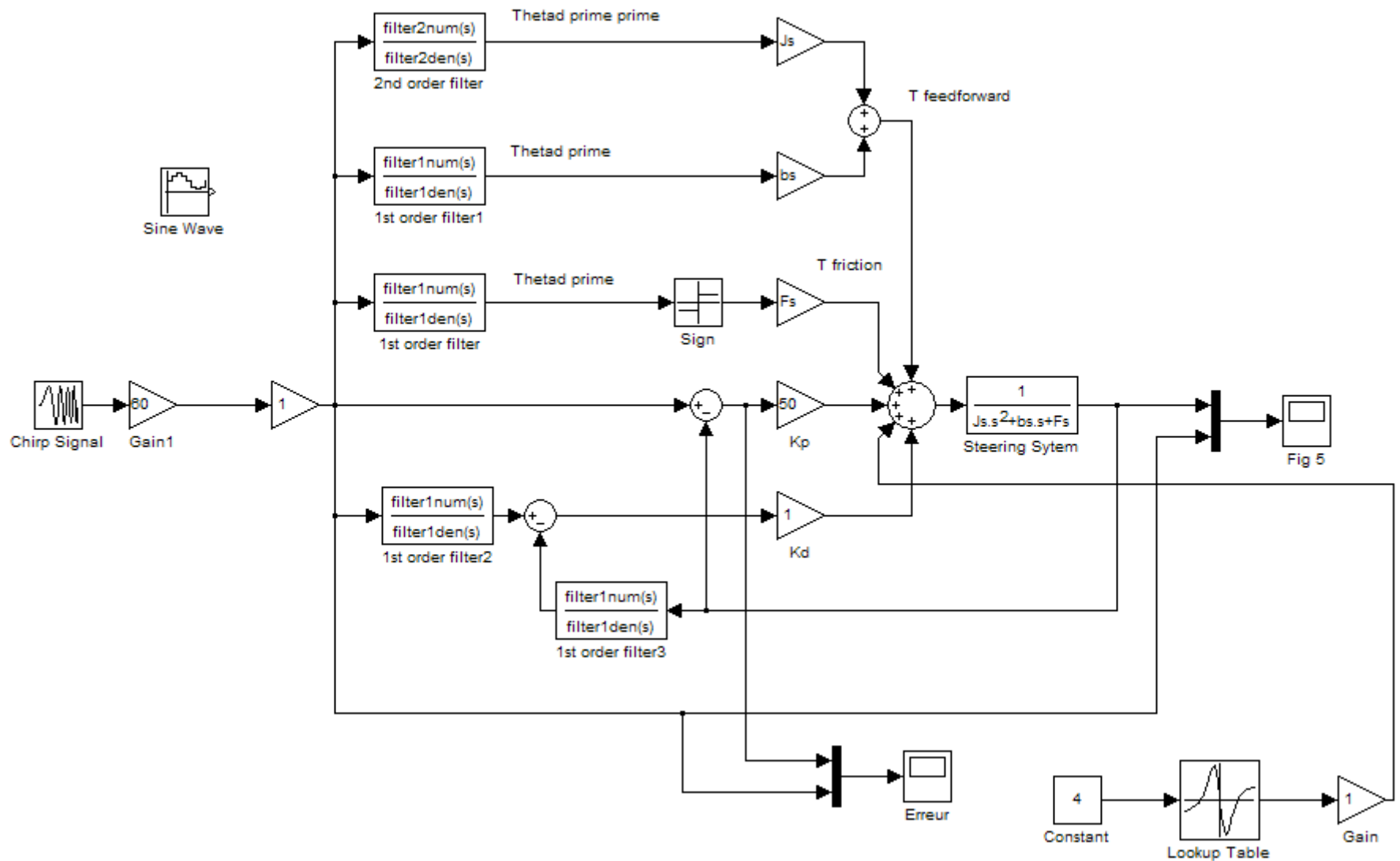


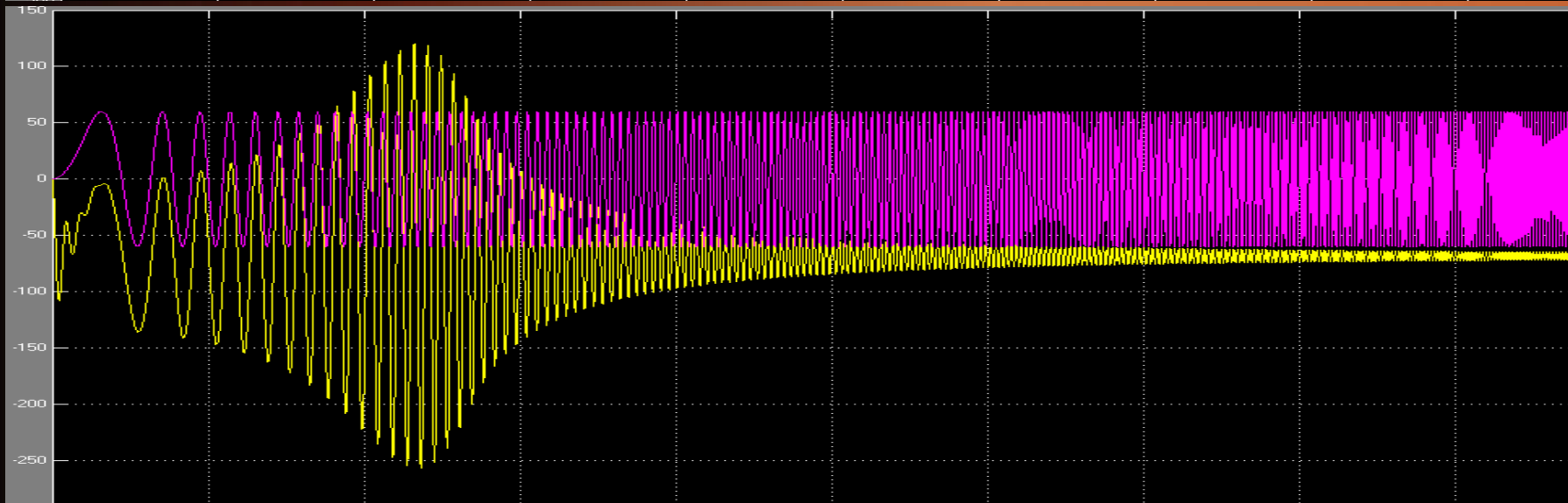
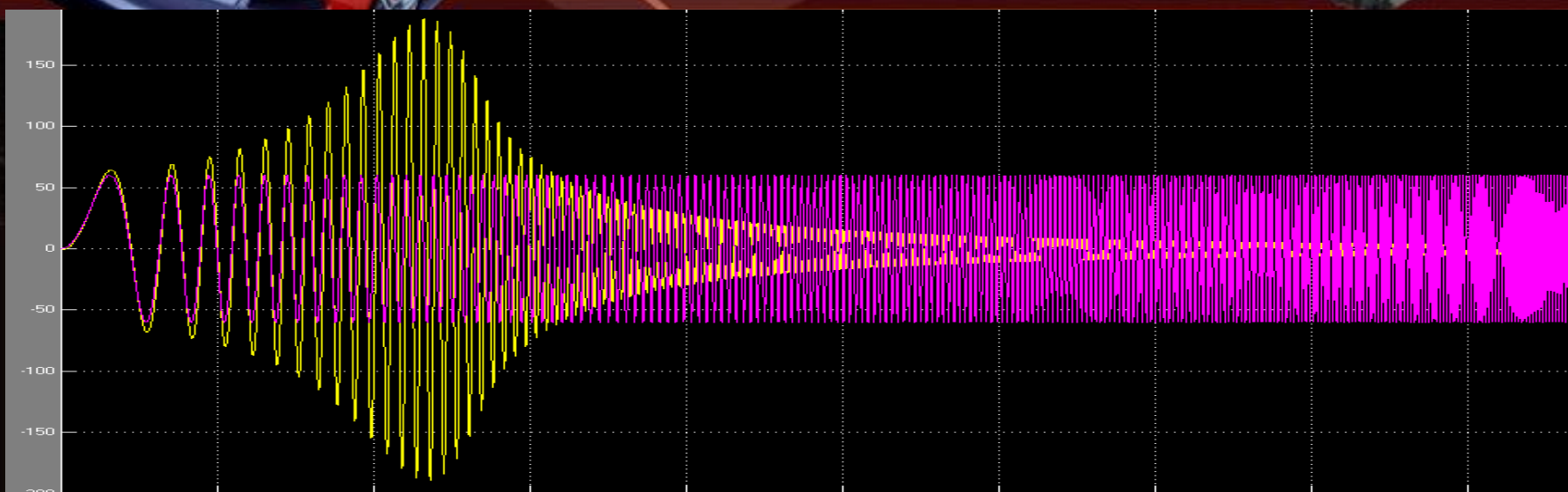


- The steering system model including the aligning moment disturbance τ_a :

$$J_s \ddot{\theta} + b_s \dot{\theta} + F_s \operatorname{sgn}(\dot{\theta}) = \tau - k_a \tau_a$$

- Where k_a is a scale factor to account for torque reduction by the steering gear







Conclusion

- Steer-By-Wire already exists in military jets and commercial airplanes.
- BMW introduced Steer-by-Wire in its 2000 prototype BMW Z22 but due to the cost involved, only implements certain components of steer-by-wire technology; they call it Active Steering.





Limitations

- Pneumatic trail, a function of slip angle, is linear for small angles
- Non-linearity problem for bigger angles
- Linearization of friction in steering block